

Deriving the inverse-elasticity Rule

- Profit function:

$$\pi(p) = (p - c)Q(p)$$

where

p is the price

c is the per-unit (marginal) cost

$Q(p)$ is the demand level when the price is p

- Set prices to maximize profit

$$p^* = \operatorname{argmax}_p (p - c)Q(p)$$

i.e., Find p^* such that $\pi(p^*)$ is the highest possible profits you can achieve.

We will use calculus to determine p^* that sets the derivative of profits equal to zero:

$$\pi'(p^*) = Q(p^*) + (p^* - c)Q'(p^*) = 0$$

$$p^* - c = -\frac{Q(p^*)}{Q'(p^*)}$$

$$\frac{p^* - c}{p^*} = -\frac{Q(p^*)}{Q'(p^*)p^*}$$

$$\frac{p^* - c}{p^*} = -\frac{1}{\frac{Q'(p^*)p^*}{Q(p^*)}}$$

$$\frac{p^* - c}{p^*} = -\frac{1}{\beta} \quad \text{Own-price elasticity}$$

- And we see that the optimal price-cost margin in % terms is equivalent to the inverse of the own-price elasticity of demand.
- Alternatively, we could mess around with that algebra and write the optimal price as follows:

$$p^* = \left(\frac{\beta}{1 + \beta} \right) c$$